A little guidance

The note inside. · Grade 2 · Parent answer key

Three gentle sessions: 15 minutes each. Understanding matters more than finishing every line.

Math · 1

15 cm longer: 24 − 9 = 15. Counting up from 9 in hops of 1, 10, and 4 is one strategy.

Math · 2

33 cm in all: 24 + 9 = 33. Split 9 into 6 and 3 to bridge through 30.

Math · 3

18 cm remains: 24 − 6 = 18. Take away 4, then 2. Each task begins with the original table lengths.

Reading · 1

Little first guessed the note was hiding inside. The key and action moved a hammer, which struck strings. The vibrating strings and soundboard made the sound as they moved.

Reading · 2

The bridge carries a string's vibration into the soundboard. The broad soundboard vibrates and moves air, which carries sound to our ears.

Writing

Accept a scene-based drawing, dictation, or 1–3 sentences. For example: The hammer taps a string and comes away. The string keeps vibrating. For pitch comparisons, keep the other string conditions the same.

Wonder first; explanations can wait. This is an acoustic piano; digital pianos usually make sound electronically through speakers. Museum models omit parts and exaggerate motion. They are not building plans or exact key/string counts. The action lets the hammer rebound while a key stays down. The bridge carries string vibration into the soundboard, which moves more air using energy from the strike. Pitch depends on vibration rate; loudness depends largely on vibration size. Compare string lengths with material, thickness, and tension unchanged: longer gives lower pitch; shorter gives higher. For the same string and vibrating length, tighter gives higher and looser gives lower. Real pianos combine length, thickness, and tension; low strings are longer and heavier. Pressing harder mainly changes loudness. Many notes use two or three strings. With the sustain pedal up, dampers return as keys are released; the highest strings often lack dampers. The story uses a middle key. Add the missing bridge or air to Little's sketch. Pictures, gentle humming, talking, and drawing are enough. Do not open a piano, touch or tighten strings, or build these models. Welcome dictation and stop whenever you have had enough.

Materials: pencil, crayons if you like, and scrap paper. Print kid pages separately from this key.